LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Quantitative displays: - Circular scale, high range long scale displays, angle of observation.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Quantitative displays: - Circular scale, high range long scale displays, angle of observation. Scale range and operating range. Straight scale digital display, dual indication and colored displays.
5. Specific Objectives	15.1.2.1 Trainee will be able to explain Quantitative displays of typical instrument.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of instrument scales using L & Ds; Safety: Handle glass faces carefully; Supervision: Instructor monitors handling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a quantitative display?
(b) Name one type of quantitative display used in aircraft instruments.
(c) Define the term 'scale range' in the context of instruments.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Circular scale characteristics • High range long scale displays • Angle of observation effects	L & Ds	PPT, IB	15 min
(b)	• Scale range and operating range definitions • Straight scale digital display features • Dual indication and colored displays	L & Ds	PPT, PDF	20 min
(c)	• Comparison of analog and digital displays • Interpretation of colored display indications • Maintenance checks for display units	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the angle of observation affects the reading on a circular scale.
(b) Differentiate between scale range and operating range with an example.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between straight scale digital displays and dual indication displays to be discussed in LP 06.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Qualitative displays: - Director displays. Head- up displays.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min